MongoDB Basic Practice Assignment Report

Database setup • CRUD operations • Query operators • Library System use case

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This report documents the MongoDB commands practised in MongoDB Shell using a local MongoDB server.

1. Introduction to MongoDB

MongoDB is a NoSQL, document-oriented database. It stores data as flexible BSON documents (a binary representation similar to JSON). Unlike a relational database that mainly uses tables and rows, MongoDB organizes data into collections and documents.

Key concepts

Concept	Meaning
Database	A container that holds collections.
Collection	A group of related documents; similar to a table in a relational database.
Document	A single record made of field-value pairs; similar to a row, but more flexible.
Field	A named data item in a document, such as name, age, or course.

2. Database and Collection Setup

A database named **studentDB** and a collection named **students** were created.

```
use studentDB\ndb.createCollection("students")
```

3. CRUD Operations

CRUD stands for Create, Read, Update, and Delete.

3.1 Create – Insert documents

```
db.students.insertOne({\n  name: "Arun", age: 22,\n  course: "MERN Stack", status: "enrolled"\n})\n\ndb.students.
```

The practice collection contained three student documents: Priya (22), Arun (23), and Kumar (24), all in MERN Stack. The final status shown for all three was completed.

3.2 Read – Find documents

```
db.students.find()\ndb.students.find({ name: "Priya" })
```

The find() method displays documents. A filter object can be supplied to return only matching documents.

3.3 Update – Update one and multiple documents

```
db.students.updateOne(\n  { name: "Priya" },\n  { $set: { age: 22 } }\n)\n\ndb.students.updateMany(\n  { status:
```

The updateMany operation returned matchedCount: 2 and modifiedCount: 2. This confirmed that two matching student documents were updated.

3.4 Delete – Delete one and delete many

```
db.students.deleteOne({ name: "Arun" })\ndb.students.deleteMany({})
```

One student document was deleted by condition. Later, deleteMany({}) returned deletedCount: 3, clearing the remaining student documents for the delete-all practice.

4. MongoDB Query Operators

Operator	Purpose	Practice query
$gt	Greater than	db.students.find({ age: { $gt: 22 } })
$lt	Less than	db.students.find({ age: { $lt: 24 } })
$in	Matches any value in a supplied list	db.students.find({ course: { $in: ["MERN Stack", "React"] } })

Operator	Purpose	Practice query
$and	All listed conditions must match	db.students.find({ $and: [{ age: { $gt: 21 } }, { status: "completed" }] })
$or	At least one condition must match	db.students.find({ $or: [{ name: "Priya" }, { age: { $gt: 23 } }] })
$exists	Checks whether a field exists	db.students.find({ course: { $exists: true } })

The operators were tested in MongoDB Shell. The observed results matched the conditions in each query.

5. Real-World Use Case – Library System

A simple Library System was created using a separate **books** collection in the same **studentDB** database. Each book document stores its title, author, category, price, and availability.

5.1 Create the collection and insert books

```
db.createCollection("books")\n\ndb.books.insertMany([\n  { title: "JavaScript Basics", author: "John Smith",\n
```

The insertMany operation successfully inserted three book documents.

5.2 Read and search books

```
db.books.find()\ndb.books.find({ category: "Programming" })
```

The category search returned JavaScript Basics and React for Beginners. Wings of Fire was not included because its category is Biography.

5.3 Update book availability

```
db.books.updateOne(\n  { title: "JavaScript Basics" },\n  { $set: { available: false } }\n)\n\ndb.books.find({ ti
```

The update operation returned matchedCount: 1 and modifiedCount: 1. A follow-up find query confirmed available: false for JavaScript Basics.

6. Conclusion

This practice introduced MongoDB database and collection setup, document insertion, reading and filtering, single and multiple document updates, deletion, and the required query operators. A Library System use case was also implemented to demonstrate how MongoDB can store and update real-world data.

Evidence note: This report summarizes the commands and results recorded during the practical session. If your instructor requires screenshots, add your own MongoDB Compass or MongoDB Shell screenshots before submission. Do not present this report as containing screenshots that are not attached.